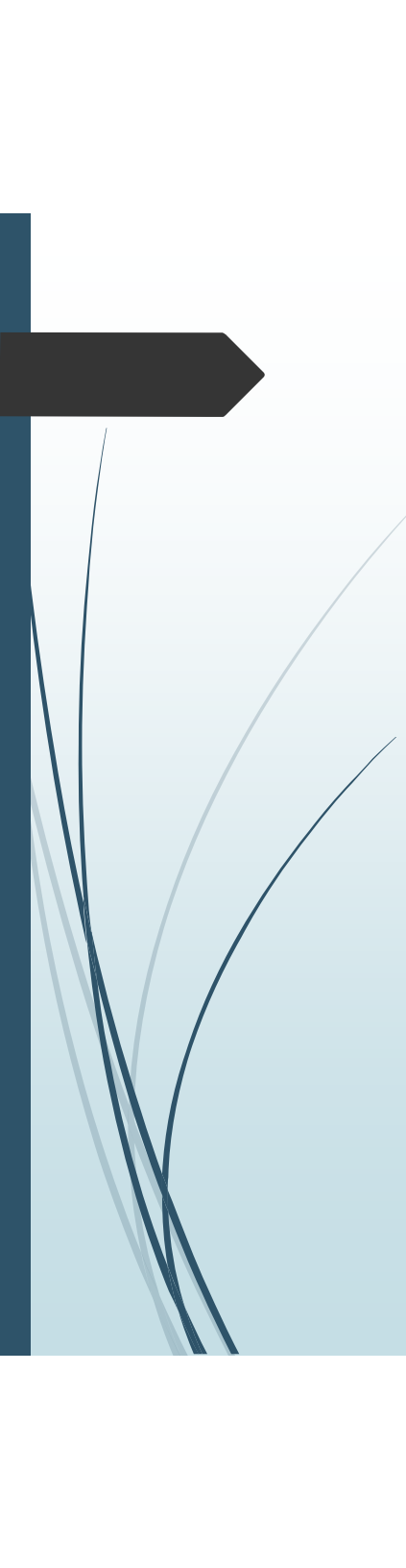




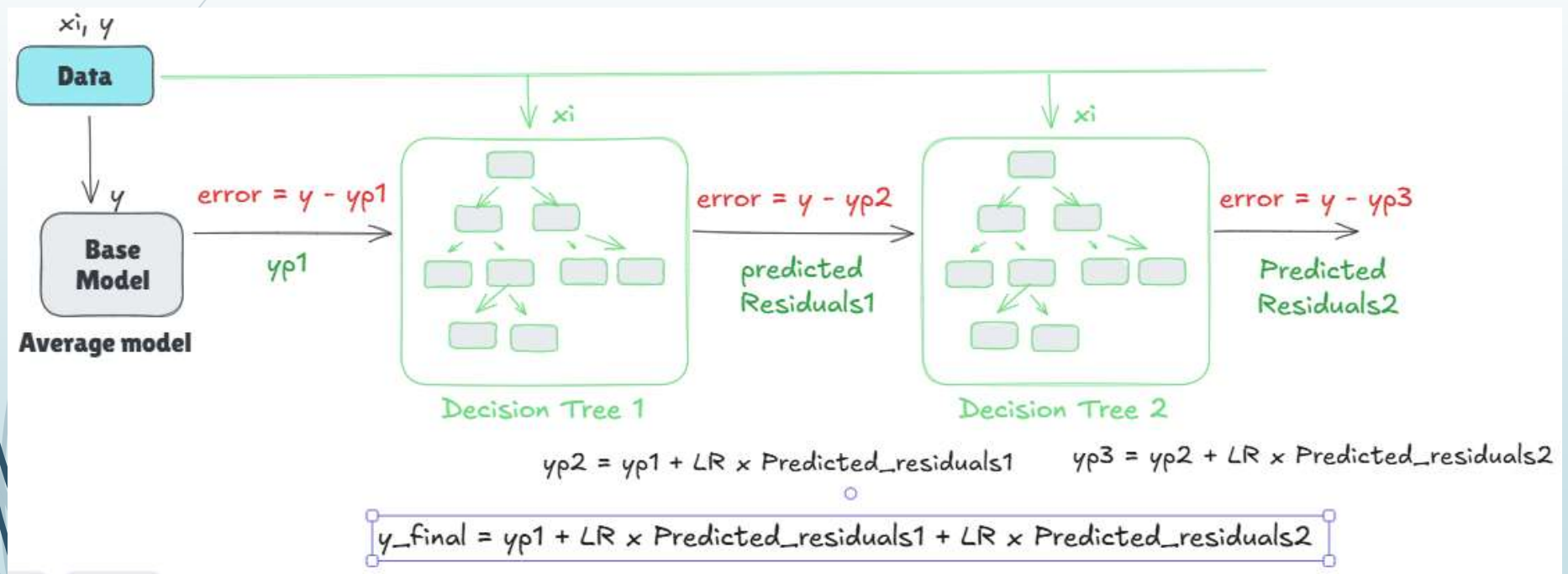
Ensembles of Decision Trees:

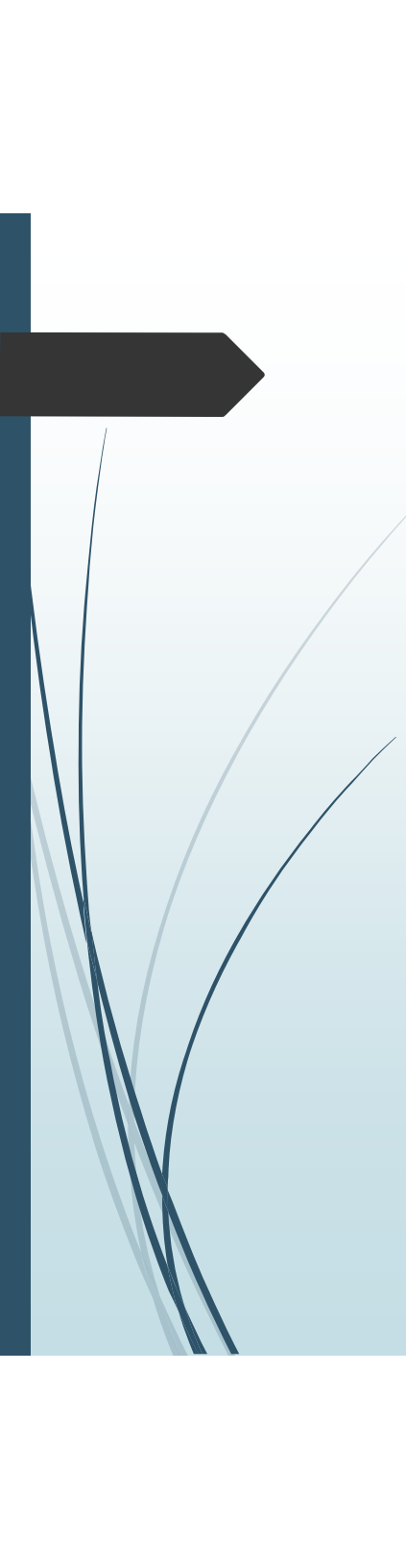
2. Gradient Boosting:

The ***gradient boosting algorithm*** is a powerful ensemble machine learning technique that builds predictive models in a sequential, stage-wise fashion. It combines multiple "weak learners," typically decision trees, to create a single, robust "strong" model that minimizes overall prediction errors by learning from the mistakes of the previous models.

Ensembles of Decision Trees:

1. Gradient Boosting:





Ensembles of Decision Trees:

Steps in Gradient Boosting:

1. We will create a base model $\rightarrow$ Average model or most frequent categories.
2. Calculate residuals (error) from average prediction and actual values.
3. Now create another model Decision Tree 1 (DT1), which will take x_i as input and residuals as target.
4. Now we have new predicted target values by DT1.
5. Now we will calculate residuals (error) again from (actual – DT1_predicted).
6. New model DT2 will fit again on input x_i and residuals as target

Ensembles of Decision Trees:

Steps

<u>area</u>	<u>Floor</u>	<u>No.of rooms</u>	<u>Price</u>		<u>area</u>	<u>Floor</u>	<u>No.of rooms</u>	<u>Price</u>	<u>Average</u>
3300	1	4	130		3300	1	4	130	108
2200	2	3	90		2200	2	3	90	108
2400	3	2	93		2400	3	2	93	108
2900	2	5	128	→	2900	2	5	128	108
1800	1	3	95		1800	1	3	95	108
1700	3	4	88		1700	3	4	88	108
3400	1	5	135		3400	1	5	135	108
		<u>area</u>	<u>Floor</u>	<u>lo.of room</u>	<u>Price</u>	<u>Average</u>	<u>Residuals</u>		
		3300	1	4	130	108	22		
		2200	2	3	90	108	-18		
		2400	3	2	93	108	-15		
		2900	2	5	128	108	20		
		1800	1	3	95	108	-13		
		1700	3	4	88	108	20		
		3400	1	5	135	108	27		

Ensembles of Decision Trees:

Steps

area	Floor	No.of rooms	Residuals		
3300	1	4	22		
2200	2	3	-18		
2400	3	2	-15		
2900	2	5	20		
1800	1	3	-13		
1700	3	4	20		New Pridicted Price
3400	1	5	27		111
					107
				DT1	107
area	Floor	lo.of room	Residuals	Predicted Residuals	
3300	1	4	22	25	110
2200	2	3	-18	-14	107
2400	3	2	-15	13	106
2900	2	5	20	24	111
1800	1	3	-13	-10	
1700	3	4	20	-17	
3400	1	5	27	31	

**New predicted price (DT1) = Average predicted value +
LR * Predicted Residuals**

Ensembles of Decision Trees:

Steps

<u>area</u>	<u>Floor</u>	<u>lo.of room</u>	<u>Price</u>	New Pridicted Price	<u>Residuals</u>
3300	1	4	130	111	20
2200	2	3	90	107	-17
2400	3	2	93	107	-14
2900	2	5	128	110	18
1800	1	3	95	107	-12
1700	3	4	88	106	-18
3400	1	5	135	111	24